

DADAL, Mongolia

WMO: 442540

Lat: 49.0222N

Lon: 111.6117E

Elev: 7156

StdP: 11.27

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-18.8	-30.2	1.3	-19.3	-26.7	1.6	-16.9	16.6	6.7	15.1	7.2	3.2	270	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	86.3	60.7	82.2	60.0	78.7	59.3	65.6	77.9	63.7	75.1	62.1	73.4	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.1	109.2	70.1	60.2	102.1	68.8	58.5	95.9	67.2	34.7	78.3	33.1	74.3	31.7	73.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.6	13.4	-27.7	92.6	4.2	4.9	-30.7	96.2	-33.2	99.0	-35.5	101.8	-38.5	105.4		
(5)			-28.2	68.5	4.3	2.4	-31.3	70.2	-33.8	71.6	-36.2	72.9	-39.3	74.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.3	-2.4	4.2	21.3	37.2	50.3	60.8	65.4	61.7	50.7	34.6	13.9	0.3		
	DBStd	25.61	8.77	10.19	11.23	9.51	8.30	6.79	4.99	6.46	7.95	9.07	11.05	9.44		
	HDD50	7489	1625	1283	890	396	98	4	0	3	85	482	1082	1541		
	HDD65	11706	2090	1703	1354	833	461	160	54	139	431	943	1532	2006		
	CDD50	1402	0	0	0	13	107	329	476	367	106	4	0	0		
	CDD65	145	0	0	0	0	5	34	66	37	3	0	0	0		
	CDH74	2040	0	0	0	6	167	573	736	491	67	0	0	0		
	CDH80	603	0	0	0	0	45	185	226	134	13	0	0	0		
	WSAvg	5.5	5.0	5.4	5.9	6.9	7.4	5.6	4.8	4.9	5.3	5.4	5.0	4.9		
	PrecAvg	14.3	0.1	0.1	0.2	0.6	1.3	2.0	4.1	3.3	1.8	0.5	0.2	0.1		
(15) Precipitation	PrecMax	21.3	0.6	0.4	0.5	1.3	3.4	5.9	6.3	7.4	4.4	1.4	0.8	0.4		
	PrecMin	10.5	0.0	0.0	0.0	0.0	0.5	0.3	0.4	0.8	0.1	0.0	0.0	0.0		
	PrecStd	2.2	0.1	0.1	0.2	0.4	0.8	1.3	1.4	1.7	1.1	0.4	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	36.2	57.5	72.7	85.3	91.6	93.4	89.2	81.6	66.1	46.2	27.8		
		MCWB	17.7	26.8	38.8	48.2	54.2	61.2	65.2	63.8	56.9	47.6	35.2	21.8		
	2%	DB	18.0	29.4	49.9	65.1	78.5	85.2	86.0	83.9	75.0	60.0	39.0	21.5		
		MCWB	14.0	22.5	35.7	43.8	51.9	59.6	62.3	61.7	55.4	43.5	31.1	17.0		
	5%	DB	13.9	25.0	43.7	59.2	73.0	80.3	81.3	79.7	70.3	54.4	34.3	17.3		
		MCWB	10.6	19.6	32.1	41.2	49.5	58.3	62.2	60.4	52.7	40.1	27.6	13.8		
	10%	DB	10.4	20.4	37.9	53.8	68.0	76.2	77.7	75.6	66.0	49.4	30.6	13.4		
		MCWB	7.7	15.9	28.8	38.3	47.7	56.9	61.6	59.3	51.2	37.5	25.0	10.7		
	0.4%	WB	18.0	27.0	39.5	49.1	55.6	65.3	68.8	67.9	60.7	48.0	35.9	21.9		
		MCDB	22.5	35.3	55.0	71.2	80.3	84.4	84.4	82.3	76.1	62.9	46.0	27.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	13.9	22.8	36.0	44.8	53.2	62.1	66.2	64.8	57.3	44.2	31.9	17.4		
		MCDB	17.6	29.2	49.2	63.6	75.2	78.4	78.0	76.6	67.2	58.6	39.5	21.3		
	5%	WB	10.7	19.6	32.4	41.7	50.8	60.2	64.6	62.6	55.0	40.5	29.1	14.0		
		MCDB	13.6	25.1	43.2	57.8	70.7	74.9	75.5	74.2	65.2	52.8	35.0	17.3		
	10%	WB	7.7	16.1	29.0	38.8	48.4	58.5	63.2	60.9	53.0	38.1	26.3	10.8		
		MCDB	10.3	20.3	37.1	52.5	65.8	72.5	73.9	71.4	64.5	48.5	31.7	13.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	20.1	21.7	23.0	26.1	24.1	21.1	22.4	23.5	21.0	17.7	14.8		
		MCDBR	18.8	23.8	27.1	31.4	35.3	31.7	28.5	29.9	30.5	28.9	20.7	16.9		
		MCWBR	15.7	18.4	16.1	16.5	16.0	12.4	10.9	12.2	14.3	15.7	13.7	14.0		
		MCDBR	18.4	23.5	26.2	29.6	31.9	26.3	22.5	23.2	24.8	26.8	20.1	16.9		
	5% WB	MCWBR	15.5	18.3	15.8	16.1	15.6	11.2	10.2	10.9	12.3	15.4	13.9	13.8		
(40) Clear-Sky Solar Irradiance	taub		0.203	0.220	0.248	0.295	0.335	0.340	0.320	0.300	0.269	0.241	0.217	0.204		
	taud		2.439	2.436	2.455	2.402	2.306	2.382	2.483	2.525	2.575	2.583	2.510	2.429		
	Ebn at Noon		291	308	312	304	293	291	296	299	301	296	283	275		
	Edh at Noon		18	24	29	34	39	36	33	30	25	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		568	944	1431	1752	1961	1940	1833	1661	1356	935	590	445		
	RadStd		21	37	54	95	114	117	111	107	95	43	26	32		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page